ELB [Elastic Load ~~Bla~~ Balancer]

→ See In actual practical world, we don't have a single server where data is stored.

→ A Load ~~Bla~~ Balancer is a device or software that distributes network traffic across multiple servers to optimize performance & capacity

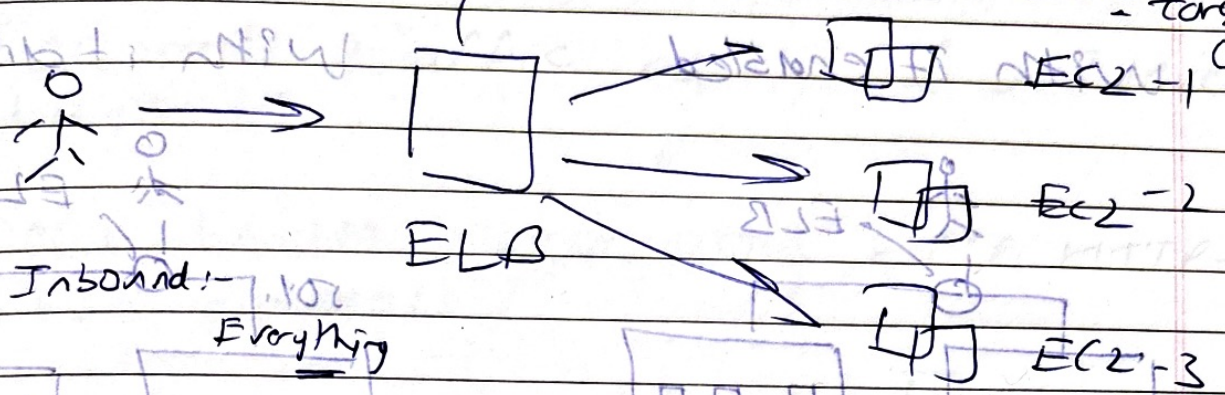
→ It ensure - high availability
- ~~fault~~ tolerance
- and a single server don't get overloaded.
- ~~scali~~ scalability.

Types of ELB

- Application LB
- Network LB
- Gatway LB
- Classic LB

very very useful

ELB



Inbound:- Everything

Inbound: sg-ELB. Best practice

- Distributing Load.
- Does health checks → If Instance not healthy → don't send traffic there.
- Provides High availability

ELB

- Application ELB
  - Forwards HTTP, HTTPS traffic
  - Application Layer-7
- Network ELB
  - Forward TCP, UDP traffic
  - Transport layer - 4
  - Really high performance
- Gateway ELB
  - Forward Network layer traffic

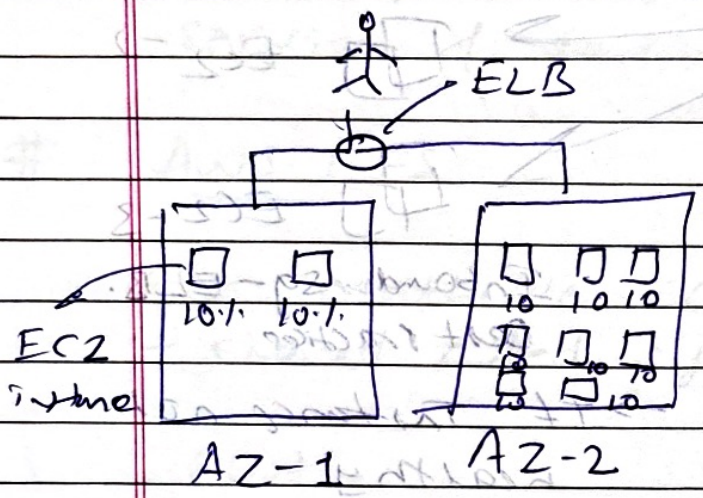
ELB vs Route 53

→ Route 53 main purpose DNS name to IP address mapping

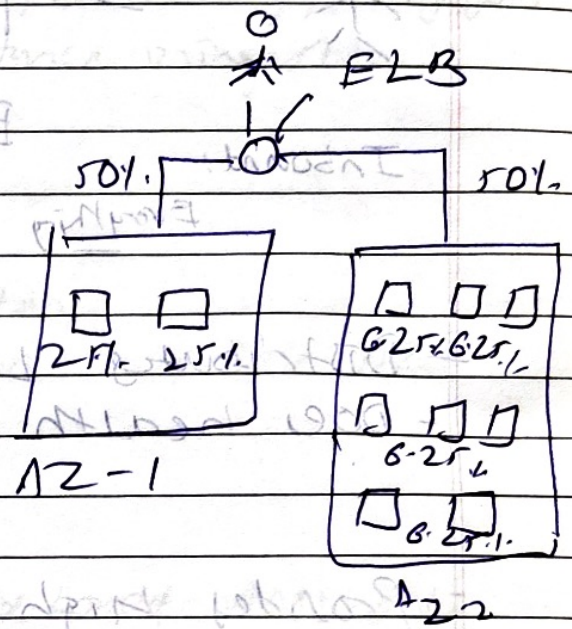
→ ELB → distribute network.

ELB Cross-zone loading:-

With it enabled



With it disabled



→ For Application ELB, cross-zone load balancing is enabled by default.

→ No extra charge

→ For NLB & GLB, you enable it so it charges you.

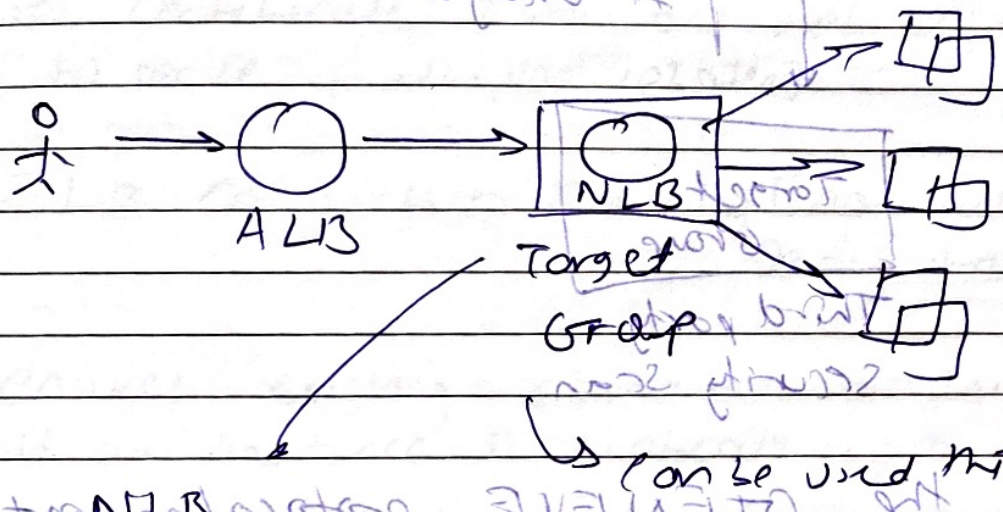
To which a ELB ~~RAM~~ distributes data to?

- EC2 instance
- IP address
- Service.

Network ELB:-

→ Forwards traffic at TCP & UDP level - Layer-4

→ Does health check using HTTP, HTTPS, TCP protocols.



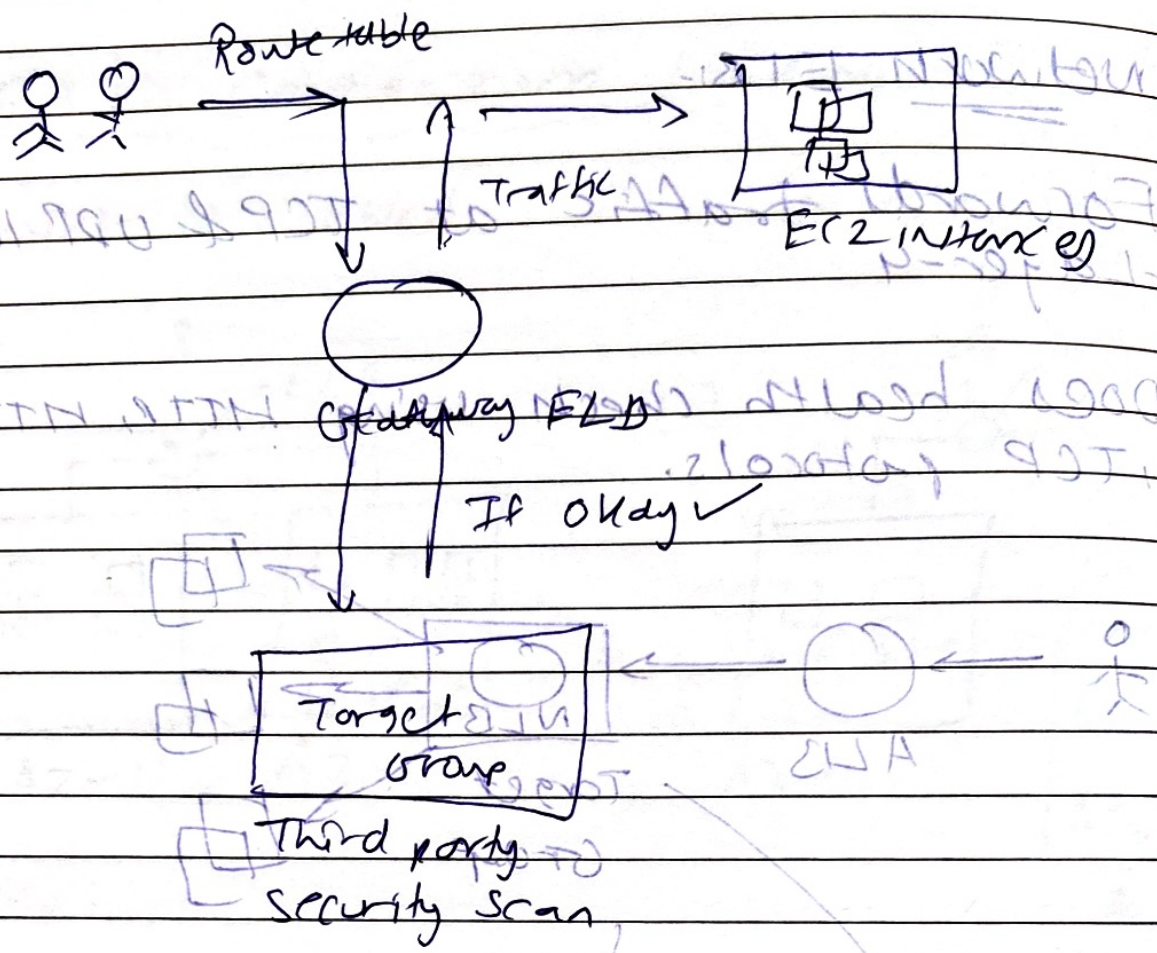
NLB gets a fixed IPV4

Gateway ELB:-

→ Forwards traffic at Network Layer

→ Deploy, scale & manage a fleet of 3rd party network virtual appliances in AWS.

Eg:- Firewall, Intrusion Detection, Deep Packet Inspection System etc



- Uses the GENEVE protocol on port 6081.

ELB Sticky sessions

→ Use cookies to preserve user data, such that when user revisits again, no need to start from ~~scratch~~ scratch.

→ Time duration → 1 day, few hours, few minutes etc

SSL: A SSL certificate allows traffic between your client & your load balancer to be ~~encr~~ encrypted in transit [in-flight encryption]

→ SSL - Secure socket layer
TLS - Transport Layer security [New version of SSL but TLS is referred to as SSL]

→ SSL - Server Name Indication [SNI]
- solves issues of handling multiple SSL certificate onto one web server [to serve multiple websites]

ELB Connection Draining [now called Deregistration Delay]

- ensures existing connection are not dropped, if an instance is unhealthy, gives a specific timeout
- Checks for a certain time, drain it if not healthy, it is smart enough that it will not send traffic to that instance in the future.
- Default - 300 seconds [Range - 1 to 3600 seconds]
- Prevents users from experiencing broken session.

Eg: Suppose you are removing an EC2 instance, without all user will be removed from the instance

But by using it, they complete their session smoothly before instance is removed.